AI MENTAL HEALTH

ANA – 1

DATASET PROFILING & SUMMARY STATS REPORT

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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1. Introduction :-

This report details the structural details of the Adult Psychiatric Morbidity Survey (APMS) 2014 provides useful data about mental health conditions among adults. The main aim of this analysis is to understand the dataset and perform basic statistical analysis on it.

2. Technology Used :-

- Language: Python — Chosen for its robust data handling and automation capabilities.
- Integrated Development Environment (IDE): Jupyter Notebook - Writing and executing code step-by-step codes, Combining code, output, and explanations.
- Core Libraries: Pandas: Primary engine for DataFrame manipulation, structural auditing, and handling 181 distinct table sheets.

3. Dataset Description :-

The dataset used in this analysis is taken from the Adult Psychiatric Morbidity Survey (APMS) 2014, conducted in England. It provides detailed statistical tables focusing on the prevalence and severity of **Common Mental Disorders (CMDs)** across various demographic groups.

	CIS-R Score Range	16-24	25-34	35-44	45-54	55-64	65-74	75+	All Adults
0	NaN	%	%	%	%	%	%	%	%
1	Men	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	0-5	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74
3	6-11	14.4	12.5	15.2	14.7	15.2	11.5	12.8	13.9
4	Under 12	90.9	84.7	84.9	86.8	85.1	92.7	94.7	87.8

Core Identification :-

- **Survey Name**: Adult Psychiatric Morbidity Survey (APMS) 2014.
- **Subject Matter**: Common Mental Disorders (CMD) such as anxiety, depression, and obsessive-compulsive disorders.
- **Primary Metric**: The **Clinical Interview Schedule – Revised (CIS-R)** score, which measures the severity of neurotic symptoms.

The dataset is divided into different categories based on severity levels:

- **0–5**: No or low symptoms
- **6–11**: Subthreshold symptoms.
- **12–17**: Mild symptoms.
- **18 or more**: Severe symptoms.

Each category contains percentage values across different age groups.

	Category	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74
13	0-5	67.1	66.3	66.4	65.3	66.3	76.6	77.6	68.6
14	6-11	15.5	16.5	15.7	16.7	16.7	13.2	14.3	15.7
15	Under 12	82.7	82.8	82.1	82.0	82.9	89.8	91.9	84.3
16	12-17	7.8	9.0	8.1	8.2	7.8	5.9	4.7	7.6
17	18 or more	9.5	8.2	9.8	9.8	9.2	4.2	3.3	8.1
18	12 or more	17.3	17.2	17.9	18.0	17.1	10.2	8.1	15.7

4. Dataset Profiling (APMS)

1. Row Count

	CIS-R Score Range	16-24	25-34	35-44	45-54	55-64	65-74	75+	All Adults
0	NaN	%	%	%	%	%	%	%	%
1	Men	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	0-5	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74
3	6-11	14.4	12.5	15.2	14.7	15.2	11.5	12.8	13.9
4	Under 12	90.9	84.7	84.9	86.8	85.1	92.7	94.7	87.8

2. Field Types (Data Types)

```
CIS-R Score Range    object
16-24                object
25-34                object
35-44                object
45-54                object
55-64                object
65-74                object
75+                  object
All Adults           object
dtype: object
```

3. Missing Values

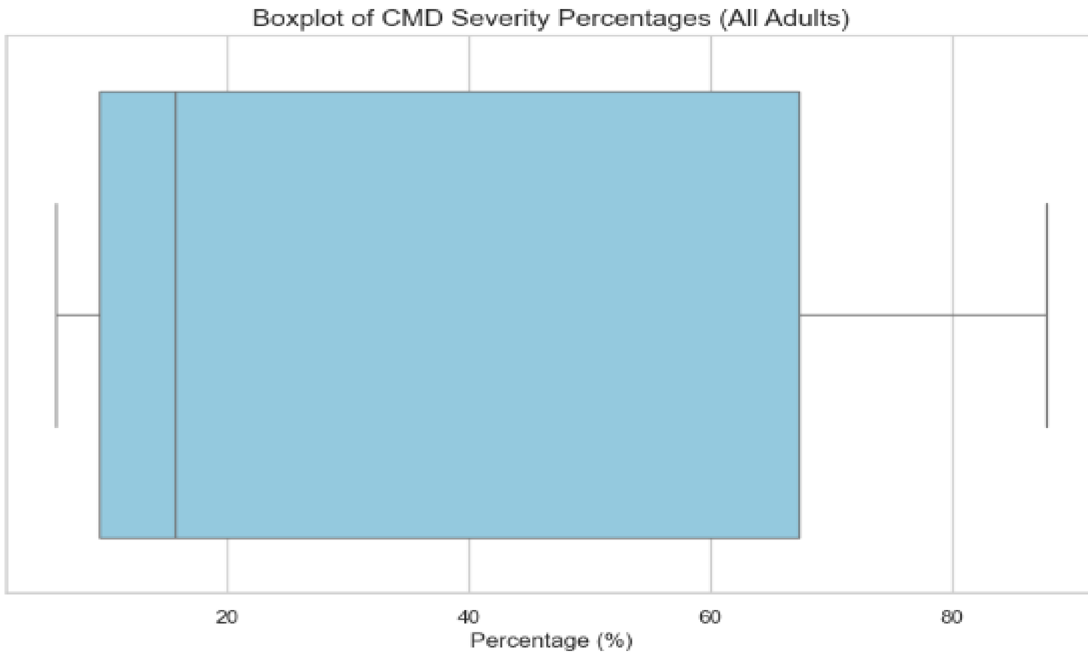
```
Row count: 29
Column count: 9
```

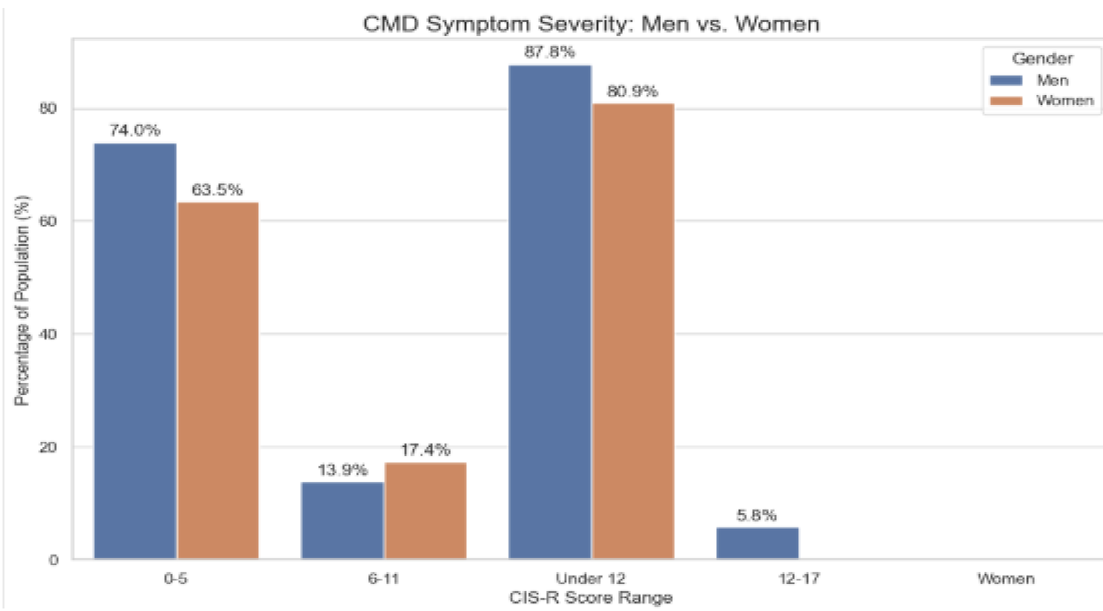
Collapse Output

```
                  Missing Count  Missing %
CIS-R Score Range             0   0.000000
16-24                         8  27.586207
25-34                         8  27.586207
35-44                         8  27.586207
45-54                         8  27.586207
55-64                         8  27.586207
65-74                         8  27.586207
75+                           8  27.586207
All Adults                    8  27.586207
```

4. Outlier Detection

```
--- Outlier Detection (All Adults) ---
   CIS-R Score Range  All Adults
22               Men      3058.0
23             Women      4488.0
24               All      7546.0
```





5. Output descriptive stats

```
--- Descriptive Statistics Table ---
count    18.000000
mean     33.338889
std      32.039298
min       5.800000
25%       9.425000
50%      15.700000
75%      67.325000
max      87.800000
Name: All Adults, dtype: float64
```

6. Conclusion

The APMS profiling workflow successfully established a reliable analytical foundation for downstream healthcare analytics and mental health trend analysis. The profiling process highlighted the importance of contextual interpretation of structural missingness and publication-oriented spreadsheet layouts commonly observed in NHS statistical datasets.

- Profiling includes:
  - Row/column counts
  - Data types
  - Missing values
  - Outlier detection